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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/345,217	06/30/2023	Si LIN	00333.0186.00US	6234

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Anova Law Group
21495 Ridgetop Circle
Suite 300
STERLING, VA 20166

EXAMINER	
KOLB, NATHANIEL J	

ART UNIT	PAPER NUMBER
2896	

NOTIFICATION DATE	DELIVERY MODE
02/06/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspto.anova@gmail.com
uspto@anovalaw.com

Office Action Summary	Application No. 18/345,217	Applicant(s) LIN et al.	
	Examiner NATHANIEL J KOLB	Art Unit 2896	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 14 Nov 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1 and 3-11 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1 and 3-11 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 14 Nov 2025 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Summary

2. Claims 1 and 3-11 are pending. Claims 1 and 3-11 are rejected herein. This is a Final Rejection as necessitated by the amendment and arguments (hereinafter "the Response") dated 14 Nov 2026.

Claim Rejections - 35 USC § 112(a)

3. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

4. Claim(s) 10 is/are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the enablement requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to enable one skilled in the art to which it pertains, or with which it is most nearly connected, to make and/or use the invention.

5. **Regarding claim 10:** The specification does not show how one obtains T_0 which is presumably the temperature inside the battery. This value has to be known to apply the formula of claim 10 to adjust the measurement of volume determined by the measurement unit. The structure or process necessary to obtain this value is not disclosed by the Applicant.

Claim Rejections - 35 USC § 112(b)

6. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

> The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

7. Claim(s) 1 and 3-11 is/are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

8. **Regarding claims 1 and 10:** Claim 1 recites that a calibrated gas volume is determined "based on…a liquid level height in the gas-washing bottle." It is unclear how this variable has any effect on the pressure involved in compensating the measured volume for temperature and pressure. The system shown in FIG. 3 of the present application must be sealed throughout and open to the atmosphere after the measurement is performed at measurement device 82. Therefore gas is produced by the battery 6 creating a pressure differential which pushes the gas through the apparatus. Therefore the only pressure that moves the gas through the system is the

pressure of the battery as it produces gas. The quantity ρgh is the hydrostatic pressure caused by the column of water in the gas-washing bottle. This pressure acts on the bottom of the gas-washing bottle 811. FIG. 3 shows that the gas never reaches this level, but is only in the headspace of gas-washing bottle 811. It is therefore unclear how the liquid level height (claim 1) or more specifically ρgh (claim 11) can have any effect on the pressure of the gas flowing through the apparatus shown in FIG. 3 and recited in claim 1.

9. **Regarding claim 10:** Claim 10 recites that "P_1 is difference between pressure in the gas-washing bottle and P_0." Claim 10 then recites that "V_1 is a gas volume of the gas-washing bottle under P_1" which seems to indicate that P_1 is the pressure inside the gas-washing bottle. It is therefore unclear what P_1 represents. Is it the absolute pressure inside the gas-washing bottle or the difference in pressure between the gas-washing bottle and P_0.

10. **Regarding claims 10:** Claim 10 is rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being incomplete for omitting essential elements, such omission amounting to a gap between the elements. See MPEP § 2172.01. The omitted elements are: a temperature sensor to determine the initial temperature of the gas. How is T_0 determined?

11. **Regarding claims 3-11:** These claims are rejected as indefinite for depending from an indefinite claim.

Response to Amendment/Arguments

12. The replacement sheet for FIG. 2 is acknowledged and the objection thereto is accordingly withdrawn.

13. The amendment to claim 11 is acknowledged and the rejection thereto under 35 U.S.C. 112(a) is accordingly withdrawn.

14. The Applicant has argued (page 8 of the Response) that claim 10 is enabled based on para. 99 of the specification as filed. This argument has been fully considered and is not persuasive. Para. 99 of the specification states that the temperature around the battery is higher than the ambient temperature. However there is nothing in the specification as to how one obtains this temperature (T_0) such that it can be used in the formula of claim 10. T_1 is measured by the measuring device 82 as stated in para. 96, but the specification is silent regarding T_0.

15. The Applicant has argued (page 11 of the Response) that the prior art does not teach "the processing device is configured to obtain the uncalibrated measured gas volume and compute a calibrated gas volume value based on…a liquid level height in the gas-washing bottle." The Examiner agrees with this statement and has removed the rejections based on prior art.

Conclusion

16. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. ROBERTSON (US 2006/0081045) teaches a gas flow meter that determines volume of gas produced (para. 47) that one skilled in the art would be motivated to use because it measures volume of gas produced while maintaining gas flow the system (para. 9). Therefore the use of a flowmeter over another method of determining the volume of produced gas is known and would be obvious to one of ordinary skill in the art.

17. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

18. A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

19. Any inquiry concerning this communication or earlier communications from the examiner should be directed to NATHANIEL J KOLB whose telephone number is (571)270-7601. The examiner can normally be reached M-F 9-5 EST.

20. Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

21. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, JESSICA HAN can be reached at (571) 272-2078. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

22. Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/NATHANIEL J KOLB/
Examiner, Art Unit 2896